

# Gemma 2 9B Compression and RX 7900 Evaluation

FAD Student Construction, Five-Fold LoRA Training, Engineering Fixes, and Hardware  
Benefit

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- ① **Context and goals** – relation to the previous RX 7900 study
- ② **Gemma 2 9B pipeline** – data, model lineage, FAD, and LoRA
- ③ **Engineering challenges** – OOM, wrong pipeline, and key collision
- ④ **Experimental evidence** – quality, parameters, VRAM, and latency
- ⑤ **Comparison and conclusion** – what improved, what did not, and why

# What This Experiment Needed to Prove

## Deployment question

Can a compressed Gemma 2 9B Student be assembled, fine-tuned, and evaluated on one 24 GiB RX 7900 XTX?

## Quality question

How much five-fold accuracy is retained relative to the merged Teacher that produced the Student?

## Hardware question

Does the 20% structural compression become lower VRAM, lower latency, or both?

## Reproducibility question

Can the training, best-adaptor selection, evaluation, benchmark, and backup chain complete without ambiguity?

**Success requires evidence across quality, compression, runtime, and engineering reliability.**

# Executive Summary

**69.86%**

**Student micro  
accuracy**

1,110 / 1,589 supported  
rows

**−4.47 pp**

**vs. merged  
Teacher**

74.32% → 69.86%

**−19.88%**

**total parameters**

9.46B → 7.58B

**−3.58 GiB**

**peak VRAM**

Approximately  
19.7–19.8% saved

## Measured interpretation

FAD delivered a clear memory reduction while preserving most Teacher quality. The quality cost is larger than in the earlier 8B compression study.

## Important limitation

Student latency remained almost identical to the Teacher (0.988–0.994× speedup). Fewer parameters did not reduce decoder depth.

**Engineering status: five folds, aggregation, inference benchmark, and verified backup completed.**

# Where the Previous RX 7900 Study Ended

| Evidence                                              | Previous result                                                         | Implication                                            |
|-------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Quality reproduction</b><br><b>8B BF16 quality</b> | RX vs. H100 fixed gaps $\leq 0.38$ pp<br>Teacher 73.63%; Student 71.87% | ROCm quality path was valid<br>1.76 pp compression gap |
| <b>Memory</b>                                         | FP4 state size 2.64–<br>$2.87\times$ smaller                            | 24 GiB deployment became feasible                      |
| <b>Runtime</b>                                        | H100 throughput 5.90–<br>$11.76\times$ higher                           | Hardware kernels still dominated                       |
| <b>RX FP4</b>                                         | W4A16; no native FP4 acceleration                                       | Memory saving $\neq$ speedup                           |

## Carry-over lesson

Compression must be reported as separate quality, memory, and speed outcomes.

## New question

Does a larger Gemma 2 9B FAD Student reproduce the same memory-first pattern?

The Gemma 2 study extends the same 1,589-row evaluation foundation.

# Scope of the Gemma 2 9B Study

## Teacher baselines

Original and merged  
Gemma 2 9B

## FAD Student

student\_best12500\_v4  
deploy bundle

## ITRI adaptation

LoRA r64, group-safe five  
folds

## RX benchmark

BF16, batch 1, four  
sequence lengths

## Included

- Five-fold supported-row accuracy
- Unsupported-row accounting
- Best-epoch selection
- Total/trainable parameters
- Latency, throughput, and VRAM
- Operational failure analysis

## Not claimed

- Native FP4 acceleration
- Reduced decoder depth
- Cross-hardware speed parity
- A universal compression optimum
- Direct equivalence with the old generation benchmark

**The fair accuracy baseline is the merged Teacher used to construct the Student.**

# Dataset and Scoring Protocol

| Item                       | Protocol                                             |
|----------------------------|------------------------------------------------------|
| <b>Evaluation design</b>   | Group-safe 5-fold English evaluation                 |
| <b>Unique rows</b>         | 1,628                                                |
| <b>Supported rows</b>      | 1,589 multiple-choice rows                           |
| <b>Unsupported rows</b>    | 39 retained for strict accounting                    |
| <b>Unsupported reasons</b> | 32 multi-answer; 6 gold absent; 1 non-classification |
| <b>Primary metric</b>      | Micro and macro accuracy on supported rows           |

**97.6%**

**rows are scorable**

1,589 / 1,628

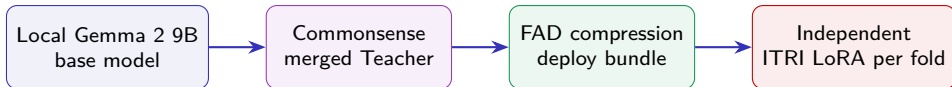
**39**

**unsupported rows**

Kept visible, not silently dropped

**Teacher and Student use exactly the same folds, prompts, and scorer.**

# Model Lineage and Fair Comparison



## Original Teacher

Reference quality before the commonsense merge.

**Micro accuracy: 74.64%**

## Merged Teacher

Fair parent baseline for the FAD Student.

**Micro accuracy: 74.32%**

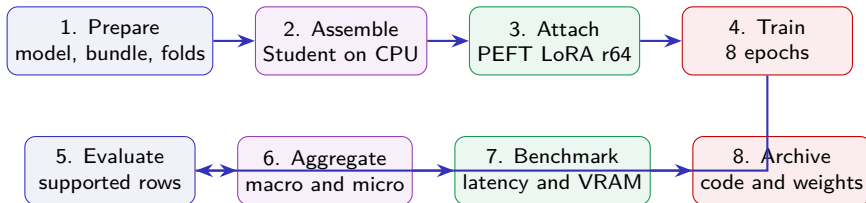
## FAD Student

Compressed child under the same protocol.

**Micro accuracy: 69.86%**

**Merge cost: 0.31 pp. Compression cost: 4.47 pp.**

# End-to-End Experimental Workflow



## Validation strategy

Smoke test → formal Fold 1 → Folds 2–5 → aggregate → benchmark. This gate prevented a 12-hour run from starting with an unverified path.

# Hardware and Software Environment

| Component            | Configuration                |
|----------------------|------------------------------|
| GPU                  | AMD Radeon RX 7900 XTX       |
| Device memory        | 23.98 GiB                    |
| Framework            | PyTorch 2.6.0 + ROCm 6.4.1   |
| Conda environment    | itri_Mlm_tf53                |
| Compute dtype        | BF16                         |
| Tokenizer/model mode | Offline local files          |
| Base-model path      | /home/ljkj/models/gemma-2-9b |

**24 GiB**

**hard deployment envelope**

The original assembly path exceeded it

**BF16**

**common comparison path**

Same dtype for Teacher and Student

**One GPU; no hidden multi-GPU offload.**

# Five-Fold LoRA Training Configuration

| Setting               | Value              | Setting             | Value |
|-----------------------|--------------------|---------------------|-------|
| Epochs                | 8                  | Batch size          | 1     |
| Gradient accumulation | 16                 | Effective batch     | 16    |
| Learning rate         | $1 \times 10^{-5}$ | Weight decay        | 0.01  |
| Warmup ratio          | 0.05               | Seed                | 44    |
| LoRA rank             | 64                 | LoRA alpha          | 64    |
| LoRA dropout          | 0.05               | Evaluation batch    | 1     |
| Max sequence length   | 768                | Max response length | 256   |

Best validation checkpoint is retained; later epochs are not automatically treated as better.

# Teacher Baselines Establish a Fair Reference

**74.64%**

**Original Teacher**

1,186 / 1,589 correct

**74.32%**

**Merged Teacher**

1,181 / 1,589 correct

**-0.31 pp**

**merge difference**

Only five fewer correct answers

## Why keep both Teachers

The original Teacher shows the best native Gemma 2 result. The merged Teacher isolates the effect of FAD because it is the direct parent of the Student.

## Interpretation

The merge did not materially degrade ITRI quality. The 4.47 pp Student gap is mainly associated with compression/adaptation, not the merge.

**Fair comparison: Merged Teacher 74.32% → FAD Student 69.86%.**

# How the FAD Student Was Constructed



## Frozen structural component

The bundle already contains internal low-rank MLP modules named `lora_up` and `lora_down`. They remain frozen during ITRI training.

## Trainable adaptation component

PEFT adds a separate LoRA r64 adapter. Only this task adapter is saved as `best_adapter` for each fold.

**Two low-rank mechanisms coexist; this later caused a checkpoint key collision.**

# Challenge 1 – Student Assembly Caused an OOM

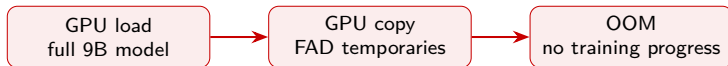
## Observed failure

HIP out of memory while deepcopy requested only 98 MiB.

- GPU total: 23.98 GiB
- PyTorch allocated: 23.72 GiB
- Free: 66 MiB
- Training steps completed: 0

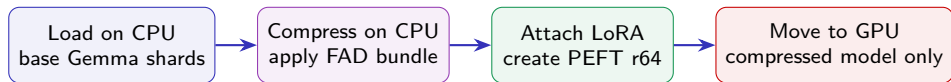
## Root cause

The original path loaded the full 9B model onto the GPU first, then assembled the shared/compressed Student there. Temporary duplication exhausted VRAM; the final Student footprint was not the problem.



**Allocator tuning could not solve a real capacity shortfall.**

## Solution 1 – Assemble on CPU, Then Move Once



**14.44 GiB**

**allocated after load**

Teacher: 18.02 GiB

**2.307%**

**trainable parameters**

174.8M / 7.58B

**Exit 0**

**smoke-test result**

Forward, backward, save,  
reload, evaluate

**The final model fit comfortably; only the original assembly peak was invalid.**

## Challenge 2 – The Trainer Imported the Wrong Pipeline

### What was wrong

`train_5fold_lora_gemma2.py` imported `newthesis_pipeline_final_llama.py` while building a Gemma 2 model.

### Why it mattered

A model-family mismatch can silently alter assembly assumptions, module names, and shared-layer behavior before any metric appears.

Llama pipeline



Gemma 2 pipeline

**Fix: point `ITRI_GEMMA_PIPELINE_DIR` to the formal Gemma 2 reproduction source.**

## Solution 2 – Validate the Correct Path Before Full Training

**16 train rows**

One optimizer update

**8 validation rows**

Best-adaptor save

**8 evaluation rows**

Reload and score

**Exit status 0**

Technical gate passed

**15.0815**

**smoke train loss**

Not a quality estimate

**9.7553**

**smoke validation loss**

Only eight examples

**4 / 8**

**smoke accuracy**

Execution check, not final  
quality

**Quality decisions were deferred to a complete formal fold.**

## Challenge 3 – FAD and PEFT Used the Same LoRA Prefix

### Failure after training

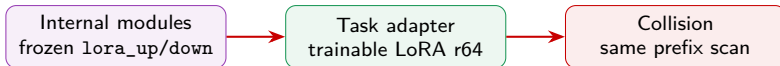
The smoke epoch finished, but reloading `best_adapter` failed with 24 unexpected keys, for example:

```
layers.25.mlp.lora_up.default.weight
```

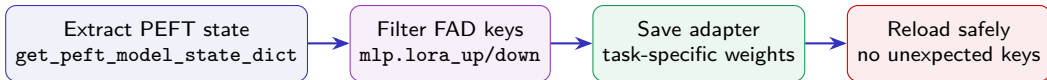
```
layers.25.mlp.lora_down.default.weight
```

### Root cause

PEFT collects keys containing the `lora_` prefix. Frozen FAD internal modules were mixed into the task-adapter checkpoint even though they were not ITRI LoRA weights.



## Solution 3 – Save Only the Trainable PEFT Adapter



**24**

**frozen FAD keys  
excluded**

Expected and logged on save

**5 / 5**

**fold adapters reload**

Best checkpoints are usable

**0**

**unexpected keys**

After the filter patch

**Filtering preserves the FAD structure while keeping PEFT checkpoints semantically clean.**

# Validation Gates and Operational Cleanup

| Gate                         | Decision rule                                                     | Observed result                             |
|------------------------------|-------------------------------------------------------------------|---------------------------------------------|
| <b>Smoke</b>                 | Train, save, and reload must pass                                 | Exit status 0                               |
| <b>Formal Fold 1</b>         | Accuracy returns to a credible range                              | 70.03%; epoch 3                             |
| <b>Folds 2–5 Aggregation</b> | Run only after Fold 1 passes<br>Five metrics and adapters present | All completed<br>Macro 69.87%; micro 69.86% |
| <b>Benchmark</b>             | Same RX protocol                                                  | Four lengths completed                      |
| <b>Archive</b>               | Copy + SHA256 verification                                        | Critical artifacts verified                 |

**97%**

**disk usage at peak**

Only 28–32 GiB free

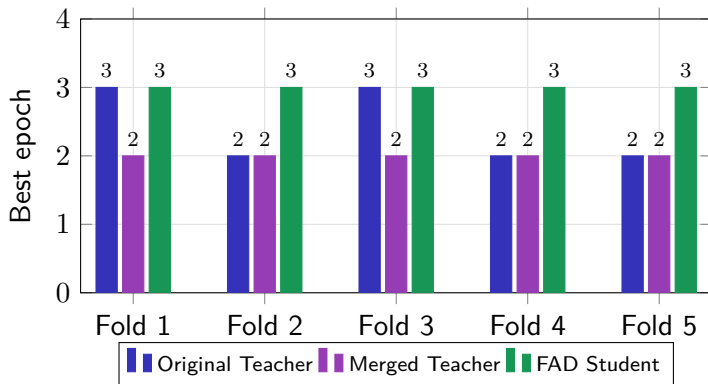
**+32 GiB**

**space reclaimed**

Removed redundant states

**Delete only after a checksum-verified off-machine backup.**

# Best-Epoch Selection Prevented Overfitting



## Student pattern

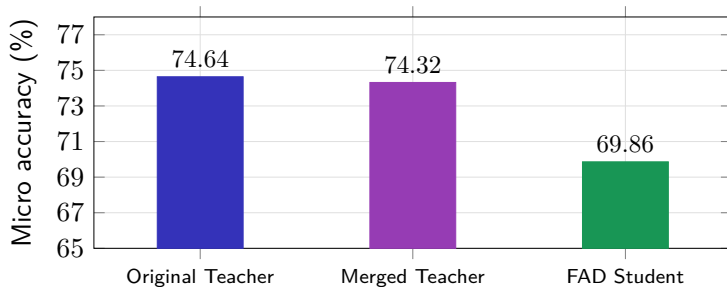
Every fold selected epoch 3.  
Later training reduced train loss but validation loss rose.

## Example: Fold 5

Epoch 8 train loss: 0.2973  
Epoch 8 val loss: 1.2168  
Saved adapter: epoch 3

Evaluation always reloads `best_adapter`, not the final epoch.

# Five-Fold Quality Results



**93.99%**

**accuracy retained**

Student / merged Teacher

**-71**

**fewer correct rows**

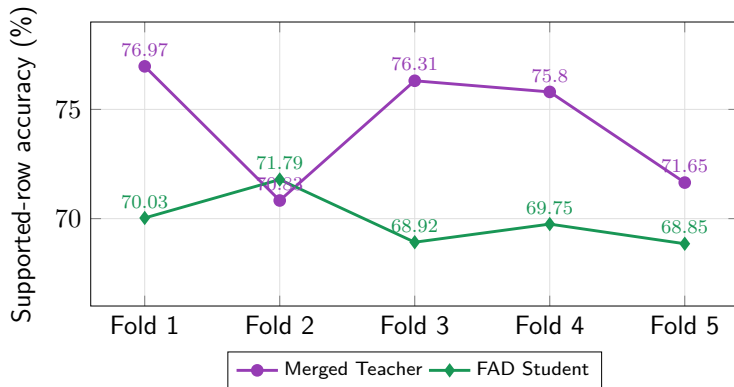
1,181 → 1,110

**$\pm 1.19$  pp**

**Student fold SD**

More stable than both Teachers

# Fold-Level Accuracy Shows One Student Win



## Fold 2

Student 71.79%

Teacher 70.83%

**Student +0.96 pp**

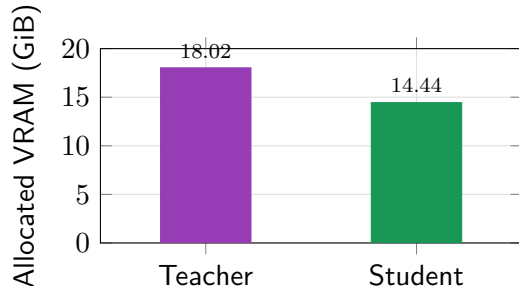
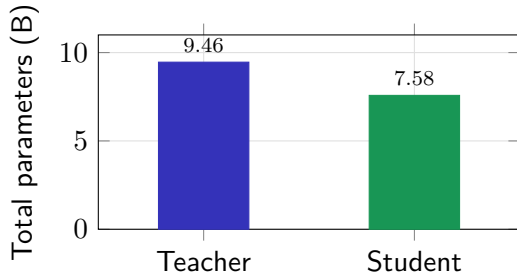
## Largest gap

Fold 3: 76.31% vs.  
68.92%

**Gap: 7.38 pp**

**Compression loss is not uniform across folds.**

# Compression Delivers a Consistent 20% Memory Reduction



**1.248×**  
parameter  
compression

9.46B → 7.58B

**−3.58 GiB**  
allocated VRAM

18.02 → 14.44 GiB

**−19.1%**  
trainable parameters

216.1M → 174.8M

# RX 7900 Inference Benchmark Protocol

| Item                    | Matched protocol                 |
|-------------------------|----------------------------------|
| <b>Hardware</b>         | Same RX 7900 XTX, GPU 0          |
| <b>Models</b>           | Merged Teacher and FAD Student   |
| <b>Precision</b>        | BF16                             |
| <b>Batch size</b>       | 1                                |
| <b>Sequence lengths</b> | 128, 256, 512, 768               |
| <b>Warmup</b>           | 5 iterations per length          |
| <b>Measured repeats</b> | 20 iterations per length         |
| <b>Reported metrics</b> | Median, P95, tokens/s, peak VRAM |

## Same inputs

### Teacher vs. Student

Controls batch and sequence length

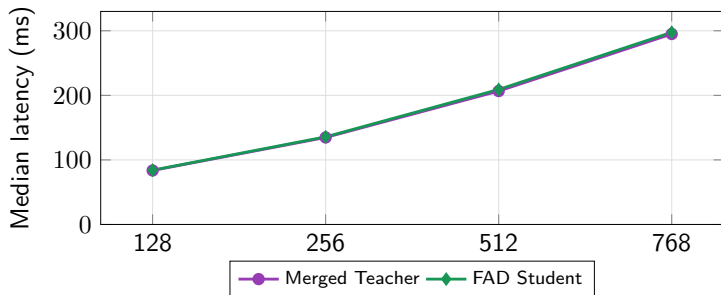
## 20 repeats

### formal measurement

After five warmup iterations

**Caveat: this forward benchmark is not numerically identical to the old 300-second generation protocol.**

# Memory Improves; Latency Does Not



## Speed

**0.988–0.994×**  
Slightly slower at every  
length

## Peak VRAM saved

**3.58 GiB**  
19.70–19.80%

## Load time

**42.98 s**  
Teacher: 21.88 s

FAD changes storage, but the full decoder path still executes.

## Compared with the Previous RX 7900 8B Study

| Metric                       | Previous 8B BF16 | Gemma 2 9B BF16 | Reading              |
|------------------------------|------------------|-----------------|----------------------|
| Teacher micro accuracy       | 73.63%           | 74.32% (merged) | Gemma +0.69 pp       |
| Student micro accuracy       | 71.87%           | 69.86%          | Gemma -2.02 pp       |
| Teacher → Student gap        | 1.76 pp          | 4.47 pp         | Higher quality cost  |
| Allocated-memory saving      | 19.28%           | 19.87%          | Nearly identical     |
| Student / Teacher throughput | 0.990×           | ≈ 0.992×        | No speedup in either |

### What reproduced

About 20% memory reduction with almost unchanged runtime.

### What changed

Gemma 2 achieved a stronger Teacher but paid a larger Student quality penalty.

Throughput ratios are within-experiment ratios because the two benchmark protocols differ.

# Measured Conclusions and Project Status

## 1. The pipeline is complete

Five folds, aggregation, adapter reload, RX benchmark, and checksum-verified backup all finished.

## 2. Quality is usable but lower

Student reaches 69.86% micro accuracy, 4.47 pp below the fair merged Teacher.

## 3. Compression benefit is real

Parameters and RX 7900 VRAM both decrease by about 20%; peak saving is 3.58 GiB.

## 4. Speed remains unchanged

Latency is 0.6–1.2% slower because structural sharing does not shorten the decoder path.

**This phase can be considered finished.**

Next work is reporting, not rerunning: document the method, show the trade-off, and preserve the artifacts.